

7.1 Waves: Transverse & Longitudinal

Question Paper

Course	CIEA Level Physics
Section	7. Waves
Topic	7.1 Waves: Transverse & Longitudinal
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A wave has a frequency of 4 GHz.

What is the period of the wave?

- A. 25 000 μs
- B. 250 ps
- C. 25 ns
- D. 2.5 ns

[1 mark]

Question 2

A light with wavelength of 600 nm has an orange colour.

What is the frequency of this light?

- A. 500 kHz
- B. 500 THz
- C. 180 GHz
- D. 180 Hz

[1 mark]

Question 3

A sound wave has a frequency of 50 Hz and a speed of 330 m s^{-1}

If two waves have a phase difference of 60° , what is the possible distance between two points on the wave?

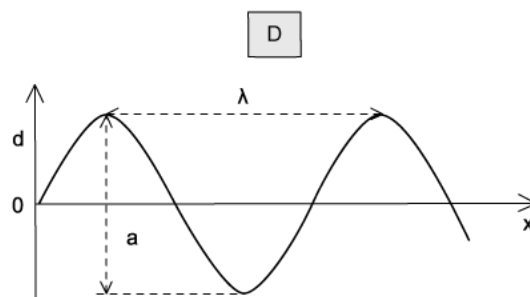
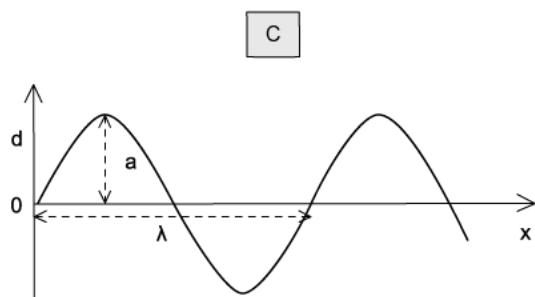
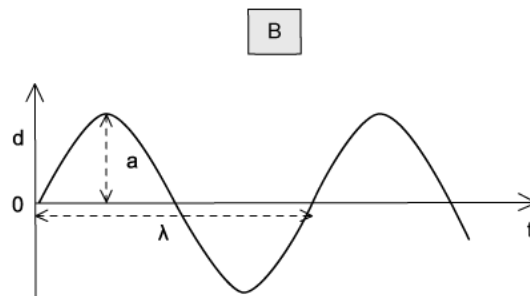
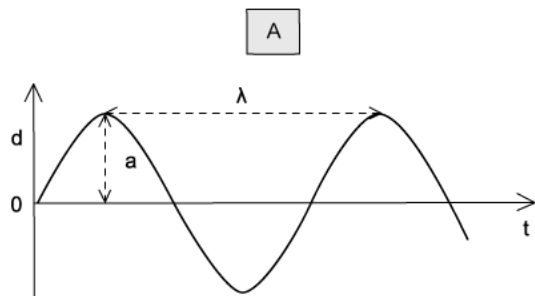
- A. 6.6 m
- B. 2.2 m
- C. 1.1 m
- D. 0.03 m

[1 mark]

Question 4

A sound wave has a displacement y at a distance x from its source at time t .

Which graph shows correctly the amplitude a and the wavelength λ of the wave?



[1 mark]

Question 5

The following are statements about waves.

- 1 Longitudinal waves can be polarised.
- 2 The amplitude of a wave is directly proportional to the energy transferred by the wave.
- 3 All electromagnetic waves travel at the same speed in a vacuum.
- 4 The frequency of infra-red light is greater than the frequency of ultra-violet light.

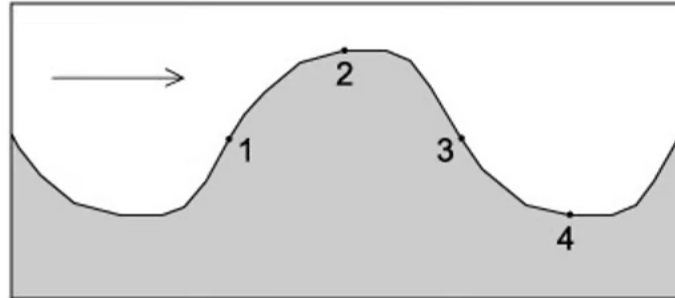
Which ones are incorrect?

- A. 1, 3 and 4
- B. 1, 2 and 4
- C. 1, 2 and 3
- D. 2, 3 and 4

[1 mark]

Question 6

The vertical cross-section diagram shows a water wave moving from left to right.



Which points on the diagram show the water moving upwards with maximum speed?

- A. 1 and 3
- B. 3 only
- C. 1 and 4
- D. 4 only

[1 mark]

Question 7

The following statements are about waves.

- 1 Waves can travel through free space
- 2 Waves can be polarised
- 3 Waves have a frequency inversely proportional to their wavelength
- 4 Waves have an intensity proportional to their amplitude

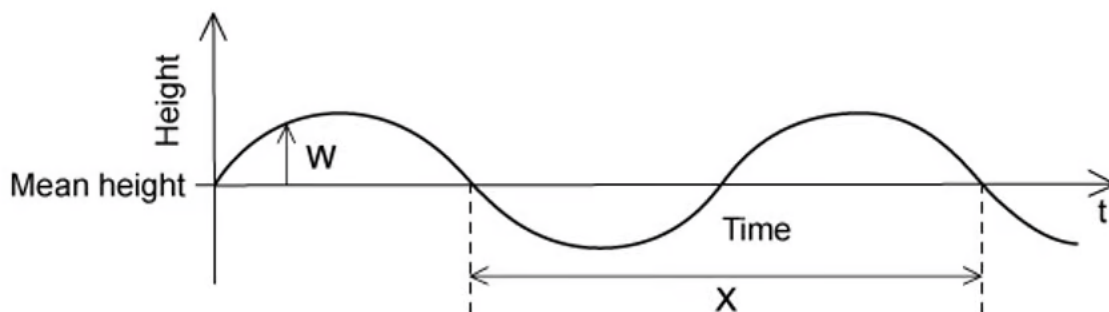
Which ones would apply to both sound and light waves?

- A. 3 only
- B. 4 only
- C. 1 and 4
- D. 1 and 2

[1 mark]

Question 8

The diagram below is a graph showing how the height of water at a point in a marina varies with time t as the waves pass the point.



What is the correct label for W and X?

	W	X
A	amplitude	period
B	displacement	wavelength
C	amplitude	wavelength
D	displacement	period

[1 mark]

Question 9

Ultraviolet waves have an order of magnitude of the frequency of the longest-wavelength of 10^x Hz.

What is the value of x ?

- A. 19
- B. 17
- C. 16
- D. 13

[1 mark]

Question 10

Which of the following statements shows that sound waves are longitudinal?

- A. sound is diffracted around corners
- B. sound is refracted as it passes from hot air to cold air
- C. sound cannot be polarised
- D. sound can be reflected from a solid surface

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- The period of a wave is $T = \frac{1}{f}$
- The question states the frequency of the wave is 4 GHz which is equal to 4×10^9 Hz
 - $T = \frac{1}{4 \times 10^9} = 2.5 \times 10^{-10}$
- 2.5×10^{-10} s is the same as 250 ps

Exam Tip: You need to be confident on converting units from milli-, micro-, nano- and pico-, especially in the topic of waves. Remember that to go from a larger unit to a smaller unit you multiply and to go from larger to smaller you divide.

2

The correct answer is **B** because:

- The speed of the wave is given by the wave equation
 - $v = f\lambda$
- The speed of light is $3.0 \times 10^8 \text{ m s}^{-1}$
- The wavelength is 600 nm or $6.0 \times 10^{-7} \text{ m}$
- The frequency is calculated by $f = \frac{v}{\lambda}$
 - So, $f = \frac{3.0 \times 10^8}{6.0 \times 10^{-7}} = 5.0 \times 10^{14} \text{ Hz}$
- There are $1 \times 10^{12} \text{ Hz}$ in 1 THz
 - Therefore, the frequency of this light is $5.0 \times 10^{14} \text{ Hz} = 500 \text{ THz}$

3

The correct answer is **C** because:

- The wavelength of the wave is
 - $\lambda = \frac{v}{f} = \frac{330}{50} = 6.6 \text{ m}$
- 60° is $\frac{60^\circ}{360^\circ} = \frac{1}{6}$ of the total wavelength:
 - $60^\circ = 6.6 \text{ m} \times \frac{1}{6} = 1.1 \text{ m}$

4

The correct answer is **C** because:

- Amplitude is the maximum displacement of a particle on the medium from its resting position
- The amplitude is shown as the arrow from the resting position to the top of the peak. This is shown in answers **A**, **B** and **C**
- The wavelength is the minimum distance between two points on the wave oscillating in phase

A & B are incorrect because the x-axis on the graph is time so this would show the period of the wave not distance

D is incorrect because the amplitude shown is the displacement of one complete wave and not from the resting position

5

The correct answer is **B** because:

- The only statement that is correct from the options listed is that all electromagnetic waves travel at the same speed in a vacuum
- Statement 1 is incorrect as longitudinal waves can't be polarised as their particles vibrate in the same direction that the wave travels
- Statement 2 is incorrect as the square of the amplitude of a wave is directly proportional to the energy transferred. This means that doubling the amplitude results in a quadrupling of the energy
- Statement 4 is incorrect as the frequency of ultra-violet light is greater than the

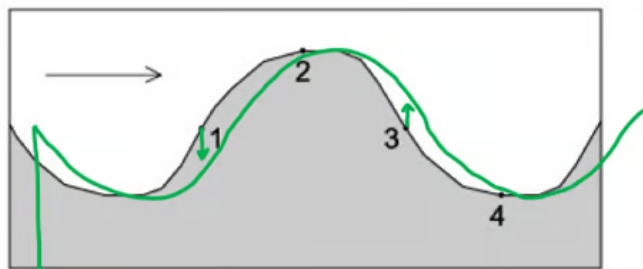
frequency of infra-red light

- Range of frequencies of infrared: $3 \times 10^{12} - 4.3 \times 10^{14}$
- Range of frequencies of ultraviolet: $7.5 \times 10^{14} - 3 \times 10^{16}$

6

The correct answer is **B** because:

- To see where the position of the particle in the next second just shift the wave in the direction of travel
- Point 3 is the only part of the wave that is moving upwards with maximum speed



From the diagram, the particle at position 1 will be moving downwards and the particle at position 3 is moving upwards. At point 4 the particle is moving up but the question states the maximum speed so **D** is incorrect. This means the correct option is **B**

7

The correct answer is **A** because:

- The only statement that applies to both light and sound is statement 3, as the relationship between frequency and wavelength is inversely proportional i.e.:
 - $f \propto \frac{1}{\lambda}$
- That means doubling the frequency halves the wavelength

B & C are incorrect because the intensity of a wave is proportional to the square of the amplitude

D is incorrect because only transverse waves can be polarised as in a longitudinal wave the particles vibrate in the same direction that the wave travels. Sound waves

are not able to travel through free space so statement 1 is incorrect

8

The correct answer is **D** because:

- The time is shown on the x-axis so **X** must be period as this is in units of time, not distance
- The amplitude of a wave is the maximum displacement (or height in this case) of the wave

A & C are incorrect because **W** is not showing the maximum displacement this is not the amplitude and just the displacement

B is incorrect because the units on the graph is time, if the graph was in m then this would be the wavelength

Exam tip: always look at the units on the graph as the first instinct for students would be to assume that **X** is the wavelength.

9

The correct answer is **C** because ultraviolet waves have a range of frequencies from $7.5 \times 10^{14} - 3 \times 10^{16}$ Hz

10

The correct answer is **C** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- Sound is a longitudinal wave
- Longitudinal waves can't be polarised as it oscillates parallel to the direction of travel
- Transverse waves can be polarised as they can oscillate in any plane

the amplitude and just the displacement

B is incorrect because the units on the graph is time, if the graph was in m then this would be the wavelength

Exam tip: always look at the units on the graph as the first instinct for students would be to assume that λ is the wavelength.

9

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B is incorrect because both transverse and longitudinal waves can be refracted as they pass from hot to cold air

D is incorrect because both transverse and longitudinal waves can be reflected from a solid surface

